DATA ITEM DESCRIPTION
Title: CONTRACTOR RELIABILITY, AVAILABILITY, MAINTAINABILITY AND TESTABILITY (RAM-T) PROGRAM PLAN

Number: DI-SESS-82362
AMSC Number: 10277
DTIC Applicable: No
Preparing Activity: MDA
Applicable Forms: N/A

Approval Date: 20220119
Limitation: N/A
GIDEP Applicable: No
Project Number: SESS-2021-035

Use/relationship: This plan describes the Contractor's Reliability, Availability, Maintainability and Testability (RAM-T) Program Plan and management process to ensure the system meets all reliability requirements. This plan documents the engineering activities and technical approach for each contracted RAM-T task including allocations, predictions, models,criticality analysis, built-in test, demonstrations, stress screening, reliability growth, stockpile reliability and failure reporting and corrective action system maintained through design, development, production, and sustainment. The principal uses of the plan are to p rovide a basic tool to manage the RAM-T program, ensure contractual compliance to the specified RAM-T requirements, and to insure that RAM-T is integrated into system engineering and throughout the system lifecycle to reduce operations and sustainment cost.

This Data Item Description (DID) contains the format, content, and intended use information for the data product that is generated by the specific task described in the contract.

Requirements:

1. Format. The RAM-T Program Plan shall be documented in a digital format using the latest Microsoft Word document format.

2. Content. The RAM-T Program Plan shall identify and describe the activities planned by the Contractor for implementation of the RAM-T program, and shall include:

    a. Scope / Objective.
    b. Applicable Documents, Standards, and Other References.
    c. The Program description of RAM-T contractual activities, the planned timing, duration, and responsible organization for each R&M and Testability activity, RAM-T requirement definition, ground rules, and process. Provide an Organizational chart that provides visibility into the resources, management and organizational structure of those responsible and accountable for the conduct of Reliability, Maintainability and BIT.
    d. The approach to perform the requisite management, design, monitoring, testing, and verification efforts for RAM-T and BIT tasks (e.g. models, allocations, predictions, fault trees, critical items lists, stress analysis, etc.), including those of the subcontractors, and methods comparing status and results to plan. Provide an integrated program schedule with all R&M and Testability test and demonstration activities aligned to major Program milestones. The schedule should include the duration of the FRACAS program and indication of Contractor or Government testing and demonstrations.
    e. Reliability growth planning in order to meet the quantitative requirement(s) including

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methods, data sources, planned growth rate, factors, test resources, growth model curve(s), etc. and showing how the tasks are integrated into the program's master schedule.

f. Explicitly include Mission Requirements Traceability Matrix showing what mission function and non-function requirement requires software capability to support vulnerability assessments and Risk Management Framework (RMF) Assessments. Relate loss of software capability to mission capability degraded or lost.
g. The Design guidelines (e.g. Mission Profile, Failure Definitions, Sneak Circuit, Worst Case, De-Rating, Thermal Stress Analysis, Mechanical Stress Analysis, Finite Element Analysis, etc.) to be used for the reliability, maintainability and BIT tasks.
h. List Government Furnished Information and Equipment necessary to support RAM-T design, development, and test activities and identify the use of RAM-T data sources (e.g.,service maintenance data collection systems).
i. Description of the intended use of the Government-Industry Data Exchange Program(GIDEP) to promote and facilitate the sharing of RAM-T information.
j. Description of tests and demonstrations planned for reliability, maintainability and BIT(e.g. Highly Accelerated Life Testing (HALT), Accelerated tests, Reliability Confidence Testing, BIT demonstrations, Maintainability demonstrations).
k. Interrelationships of reliability, maintainability, and BIT tasks, including how the resulting data will be integrated with system safety, human factors, and other system oriented tasks (e.g., design and life cycle cost analyses, level of repair analyses, availabilityanalyses, trade-off studies, etc.).
l. The appendices listed below shall appear in the RAM-T Program plan, and should detail ground rules and assumptions, processes, models, etc., and shalluse the referenced guidance:

RAM-T Appendices	Guidance
Appendix A – Testability Plan	MIL-HDBK-2165
Appendix B – Software Reliability Program Plan	See Section 3
Appendix C – Reliability Growth Plan	MIL-HDBK-189C
Appendix D – Reliability Assessment Plan	See Section 4 MIL-HDBK-189C
Appendix E – Closed Loop FRACAS Plan	DI-SESS-80255A MIL-HDBK-2155
Appendix F – Screen Strength Plan	MDA HALT/HASS HDBK MIL-HDBK-344A MIL-HDBK-2164A
Appendix G – FMECA Plan	DI-SESS-81495B
Appendix H – Stockpile Reliability Plan	See Section 5

3. Software Reliability Program Plan. Plan shall include:

a. Scope / Objective.
b. Applicable Documents, Standards, and Other References.
c. Identify the roles and responsibilities associated with development of reliable software including the role of software developers, system engineers, testability engineers, and R&M engineers.
d. Schedules, progress reporting, deliverables, resources of the software reliability

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engineering effort.

e. The plan for the design for software reliability achievement in accordance with contract requirements.
f. The plan for meeting software reliability allocation requirements including the technical approach, software FEMA, model, schedule, deliverables, and integration with the system requirements baseline.
g. The plan for verification of software reliability including metrics, randomized level testing and system level simulation.
h. The plan for verification, validation, and accreditation for models and simulations.

4. Reliability Assessment Plan. Plan shall include:

a. Scope / Objective.
b. Applicable Documents, Standards, and Other References.
c. Requirements definition, ground rules, processes including model description and verification methods to meet the quantitative requirements.
d. Plans for development of a probabilistic reliability model (e.g. National Aeronautics and Space Administration (NASA) Probabilistic Risk Assessment (PRA) model) utilizing event sequence diagrams utilizing event sequence diagrams (ESDs), fault trees, test data, failure distributions, etc., to be used for system and component Reliability assessments, projections, and growth analysis.
e. Analysis factors such as dormancy failure modes and rates, aging, pre-launch residual failures, etc.
f. Plans for a model that calculates Reliability, Availability, and Maintainability from operational system operating times and corrective and preventative maintenance downtimes.

5. Stockpile Reliability Program Plan. Plan shall include:

a. Scope/ Objective.
b. Applicable Documents, Standards, and Other References.
c. Ground rules, roles and responsibilities, process, life cycle alignment, schedule, testplanning, method of data collection and trending.
d. Overall plan for demonstrating service and reliability through life cycle, to include:
    1. Initial service life assessment reports on expected component service life based on FMECA and other information available to developers;
    2. Identification of limited life items;
    3. Strategy for demonstrating assessed service life, including service life extension for ordnance and one-shots;
    4. Identification of initial hardware and quantity requirements (lot requirements for one-shots) for service life validation of limited life items;
    5. Activities and schedules for periodic inspection, surveillance and testing of each limited life item.
e. Method of reporting including analysis results, data delivery, limited life items identification, performance parameters measuring degradation, and system operating life compliance.

End of DI-SESS-82362

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